WEEK-1

1. a) Load VADeaths (Death Rates in Virginia) dataset in R and visualize the data using different histograms.

AIM:

To visualize the death rates in virginia dataset using histograms.

DESCRIPTION:

The *VADeaths* dataset contains death rates for four demographic groups in Virginia: **Rural Male, Rural Female, Urban Male, and Urban Female**. After converting the dataset into a data frame, histograms were plotted for each group using a 2×2 layout. All four histograms were displayed simultaneously in a single plotting window.

Each histogram shows the **frequency distribution** of death rates for the respective group. From the plots, we observe that:

- **Urban Male** death rates are the highest among all groups.
- **Urban Female** death rates are lower than Urban Male but still higher than Rural groups.
- **Rural Male** death rates are higher than Rural Female but lower than Urban groups.
- **Rural Female** death rates are the lowest overall.

These histograms help compare the distribution patterns across demographic groups and clearly show that **urban populations have higher death rates than rural populations**, and **males consistently have higher death rates than females**.

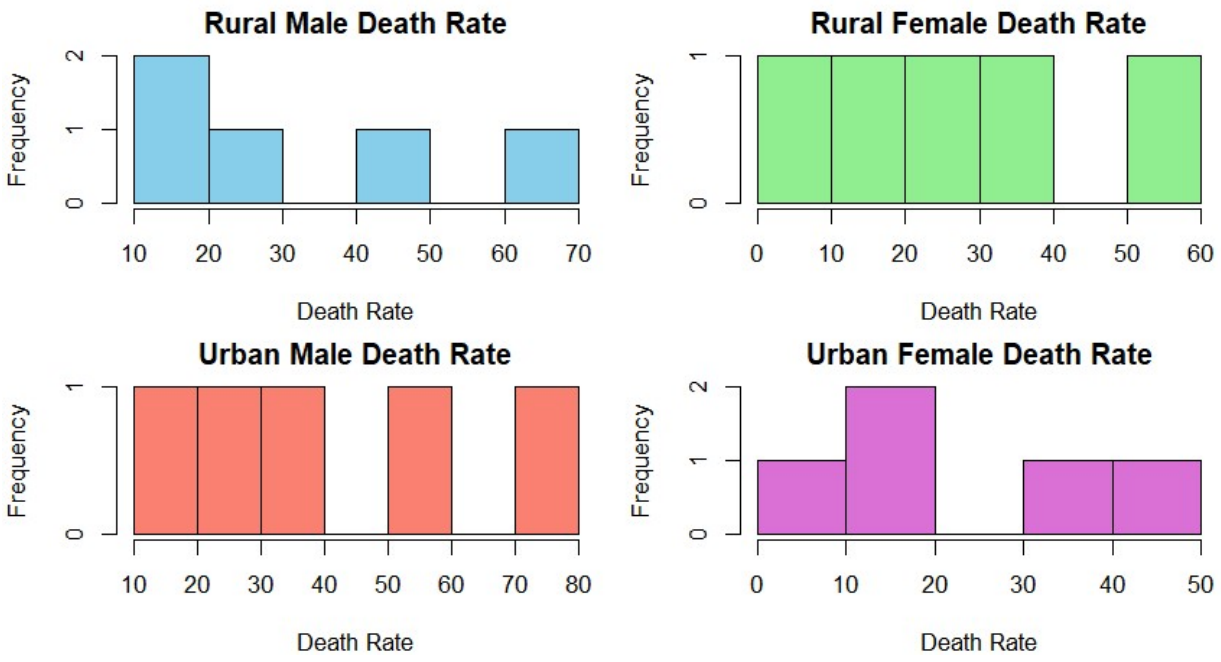
PROGRAM:

```
# Load VADeaths dataset (built-in)
data("VADeaths")

# Display the dataset
print("VADeaths Dataset:")
print(VADeaths)

# Convert matrix to data frame for easier manipulation
va_df <- as.data.frame(VADeaths)

# Optional: View structure
str(va_df)

# Visualize each column using histograms
```

```
dev.off()
par(mfrow = c(2, 2),mar=c(4,4,2,1))  # 2x2 plotting layout where mf is multi figure layout by rows

# Histogram for Rural Male
hist(va_df$`Rural Male`,
     main = "Rural Male Death Rate",
     xlab = "Death Rate",
     col = "skyblue",
     border = "black")

# Histogram for Rural Female
hist(va_df$`Rural Female`,
     main = "Rural Female Death Rate",
     xlab = "Death Rate",
     col = "lightgreen",
     border = "black")

# Histogram for Urban Male
hist(va_df$`Urban Male`,
     main = "Urban Male Death Rate",
     xlab = "Death Rate",
     col = "salmon",
     border = "black")

# Histogram for Urban Female
hist(va_df$`Urban Female`,
     main = "Urban Female Death Rate",
     xlab = "Death Rate",
     col = "orchid",
     border = "black")

# Reset layout
par(mfrow = c(1, 1))
```

OUTPUT:

	Rural Male	Rural Female	Urban Male	Urban Female
50-54	11.7	8.7	15.4	8.4
55-59	18.1	11.7	24.3	13.6
60-64	26.9	20.3	37.0	19.3
65-69	41.0	30.9	54.6	35.1
70-74	66.0	54.3	71.1	50.0


Rural Male Death Rate
Frequency
0
1
2
10
20
30
40
50
60
70
Death Rate
Rural Female Death Rate
Frequency
0
1
0
10
20
30
40
50
60
Death Rate
Urban Male Death Rate
Frequency
0
1
10
20
30
40
50
60
70
80
Death Rate
Urban Female Death Rate
Frequency
0
1
2
0
10
20
30
40
50
Death Rate

b) Load air quality dataset in R and visualize La Guardia Airport‟s daily maximum temperature using histogram.

AIM:

To visualize the maximum temperature in La Guardia Airport‟s using histogram.

DESCRIPTION:

The air quality dataset is a built-in dataset in R that contains daily air quality measurements in New York City. It was collected at La Guardia Airport from May to September 1973. The dataset has 153 observations (rows) and 6 variables (columns) such as Ozone, Solar.R, Wind, Temp, Month, and Day.

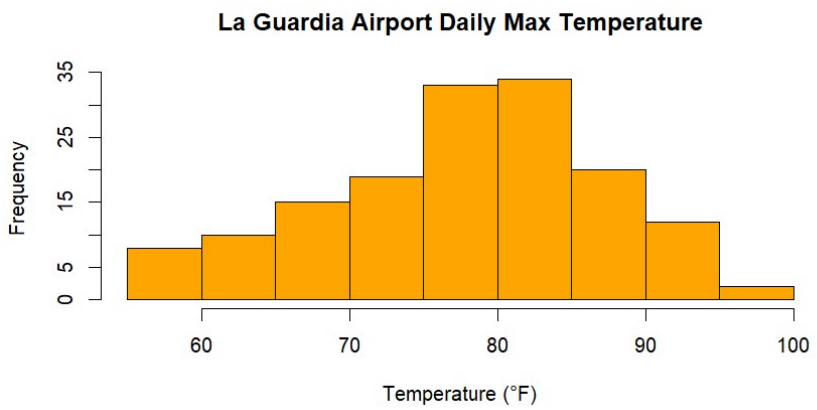
A histogram is created for the daily maximum temperature (Temp) recorded at La Guardia Airport. The histogram visually represents the frequency distribution of temperature values. The plot includes a title, labeled x-axis, and uses an orange color for better clarity. From the histogram, we can observe the temperature range and how frequently different temperature levels occur. This helps in understanding the overall temperature pattern for the given period.

PROGRAM:

```
# Load airquality dataset (built-in)
data("airquality")

# Display the first few rows
head(airquality)

# Structure of the dataset
str(airquality)

# Histogram for daily maximum temperature (Temp)
hist(airquality$Temp,
     main = "La Guardia Airport Daily Max Temperature",
     xlab = "Temperature (°F)",
     col = "orange",
     border = "black")
```

OUTPUT:

```
Ozone Solar.R Wind Temp Month Day
1    41     190  7.4   67     5   1
2    36     118  8.0   72     5   2
3    12     149 12.6   74     5   3
4    18     313 11.5   62     5   4
5    NA      NA 14.3   56     5   5
6    28      NA 14.9   66     5   6

'data.frame':   153 obs. of  6 variables:
 $ Ozone  : int  41 36 12 18 NA 28 23 19 8 NA ...
 $ Solar.R: int  190 118 149 313 NA NA 299 99 19 194 ...
 $ Wind   : num  7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
 $ Temp   : int  67 72 74 62 56 66 65 59 61 69 ...
 $ Month  : int  5 5 5 5 5 5 5 5 5 5 ...
 $ Day    : int  1 2 3 4 5 6 7 8 9 10 ...
```



WEEK – 2

2. Load AirPassengers dataset in R and visualize the data using line chart that shows increase in air passengers over given time period.

AIM:

To visualize the increase in air passengers over a given time period using line chart.

DESCRIPTION:

The built-in AirPassengers dataset is loaded and analyzed. This dataset contains the monthly total number of international airline passengers (in thousands) from 1949 to 1960.

First, the summary() function is used to display basic statistics of the dataset, including minimum, maximum, median, mean, and quartiles, which helps understand the range and distribution of passenger numbers. The paste() function concatenates text strings in R. The class() function confirms that AirPassengers is a time-series object, meaning it is ordered by time and can be used for time-based analysis.

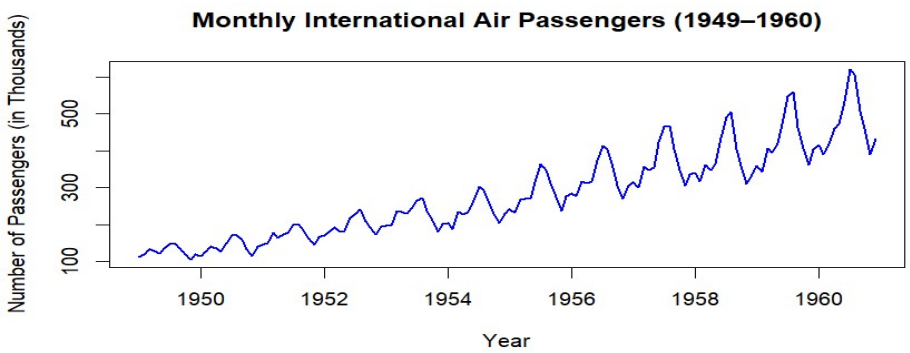
A line chart is plotted to visualize the trend in monthly passenger numbers over the years. This visualization helps in understanding **trends, patterns, and seasonal variations** in the airline passenger data over the 12-year period.

PROGRAM:

```
# Step 1: Load the dataset
data("AirPassengers")

# Step 2: Display basic details of the dataset
print("Summary of AirPassengers dataset:")
summary(AirPassengers)

# Step 3: Check the class (time-series object)
print(paste("Class of dataset:", class(AirPassengers)))

# Step 4: Visualize using a line chart
plot(AirPassengers,
     main = "Monthly International Air Passengers (1949–1960)",
     xlab = "Year",
     ylab = "Number of Passengers (in Thousands)",
     col = "blue",
     type = "l",   # Line chart
     lwd = 2)      # Line width
```

OUTPUT:

```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 104.0   180.0   265.5   280.3   360.5   622.0

"Class of dataset: ts"
```



WEEK – 3

3. a) Load iris dataset in R, visualize the data using different Bar Charts and also demonstrate the use of stacked plots.

AIM:

To visualize the iris dataset using different bar charts.

DESCRIPTION:

The program performs exploratory data analysis and visualization on the built-in **iris** dataset in R. The dataset is first loaded and displayed using head() to observe the first few records. A statistical summary of all variables is generated using summary(), which provides insights into minimum, maximum, mean, and quartile values of the flower measurements. A **bar chart** is created to show the **number of samples for each Iris species**, helping to visually understand the distribution of species in the dataset.The program then calculates the **mean Sepal Length for each species** using the aggregate() function, and the results are visualized through another bar plot.

To analyze Sepal Length patterns further, the data is grouped into four ranges using cut() and a **stacked bar plot** is generated. This plot shows how samples of each species are distributed across different Sepal Length ranges.

Overall, the program provides a detailed visual and statistical exploration of the iris dataset using data summaries, bar charts, and grouped visualizations.

PROGRAM:

```
# Step 1: Load the dataset
data("iris")

# Step 2: View the first few rows
head(iris)

# Step 3: Summary of the dataset
summary(iris)

# Step 4: Create a bar chart showing number of samples per species
species_count <- table(iris$Species)

barplot(species_count,
        main = "Number of Iris Samples per Species",
        xlab = "Species",
        ylab = "Count",
        col = c("skyblue", "salmon", "lightgreen"),
```

```
        border = "black")
```

 # Step 5: Mean Sepal Length per Species (Base R calculation)

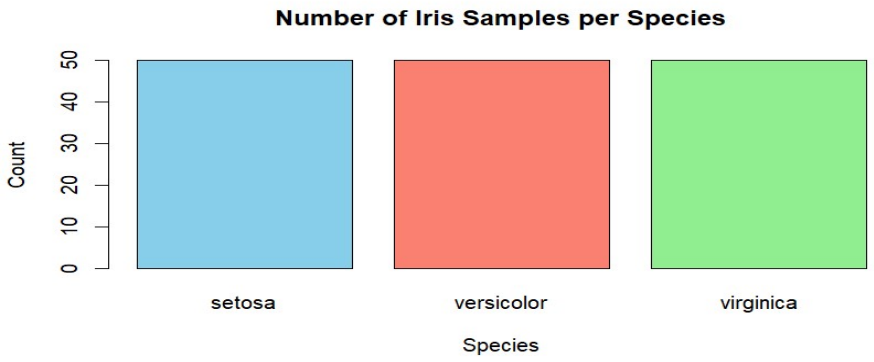
```
mean_sepal <- aggregate(Sepal.Length ~ Species, data = iris, FUN = mean)

barplot(mean_sepal$Sepal.Length,
        names.arg = mean_sepal$Species,
        main = "Mean Sepal Length by Species",
        xlab = "Species",
        ylab = "Mean Sepal Length",
        col = "orange")
```

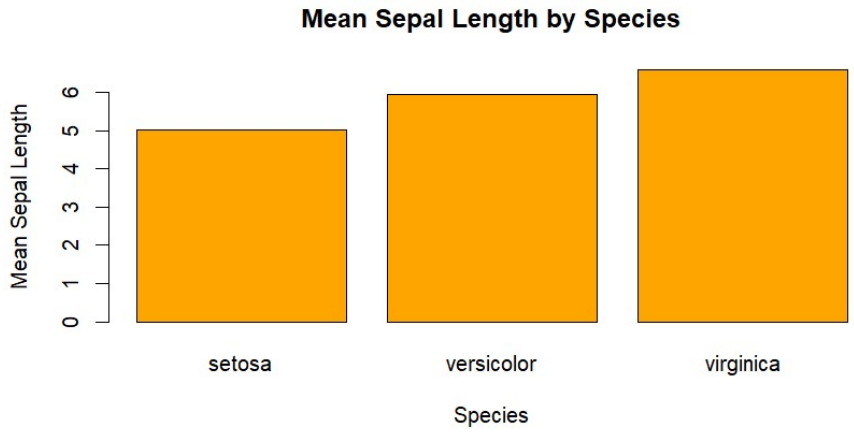
 # Step 6: Stacked Bar Plot – Counts of Species by Rounded Sepal Length

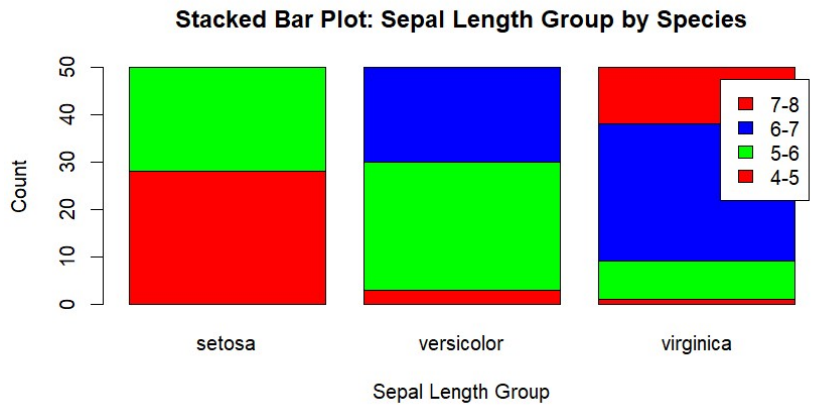
```
iris$SepalGroup <- cut(
  iris$Sepal.Length,
  breaks = c(4, 5, 6, 7, 8),
  labels = c("4-5", "5-6", "6-7", "7-8")
)

stack_data <- table(iris$SepalGroup, iris$Species)

barplot(stack_data,
        main = "Stacked Bar Plot: Sepal Length Group by Species",
        xlab = "Sepal Length Group",
        ylab = "Count",
        col = c("red", "green", "blue"),
        legend = rownames(stack_data),
        beside = FALSE)
```

OUTPUT:

```
Sepal.Length Sepal.Width Petal.Length Petal.Width
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
       Species
 setosa     :50
 versicolor:50
 virginica :50
Sepal.Length    Sepal.Width     Petal.Length    Petal.Width          Species
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100   setosa    :50
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300   versicolor:50
 Median :5.800   Median :3.000   Median :4.350   Median :1.300   virginica :50
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
```


Number of Iris Samples per Species
Count
0
10
20
30
40
50
setosa
versicolor
virginica
Species


Mean Sepal Length by Species
Mean Sepal Length
0
1
2
3
4
5
6
setosa
versicolor
virginica
Species


Stacked Bar Plot: Sepal Length Group by Species
Count
0
10
20
30
40
50
7-8
6-7
5-6
4-5
setosa
versicolor
virginica
Sepal Length Group

b) Load air quality dataset in R and visualize ozone concentration in air.

AIM:

To visualize ozone concentration in air by loading air quality dataset.

DESCRIPTION:

The program performs exploratory data analysis and visualization on the built-in **airquality** dataset in R, which contains daily air quality measurements in New York. The dataset is loaded and the first few rows are displayed using head() to understand the structure and types of variables present. A statistical summary of all variables is generated using summary(), giving information on minimum, maximum, mean, and missing values.

A **histogram** of the *Ozone* variable is created to visualize the distribution of ozone concentration in the atmosphere. This helps in identifying common ranges and overall frequency patterns. A **boxplot** of *Ozone* values is used to study the spread of the data, detect skewness, and identify possible outliers. A **line plot** is generated to show how ozone levels vary over time (day index). Each data point is also marked using points() to highlight individual readings, making the trend clearer.

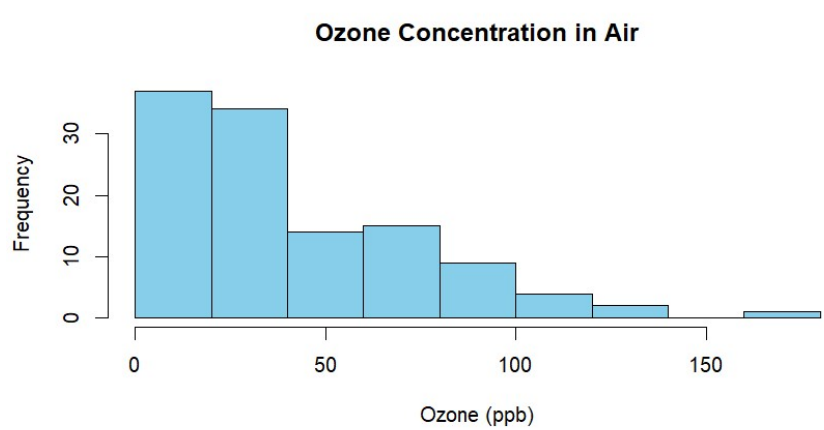
PROGRAM:

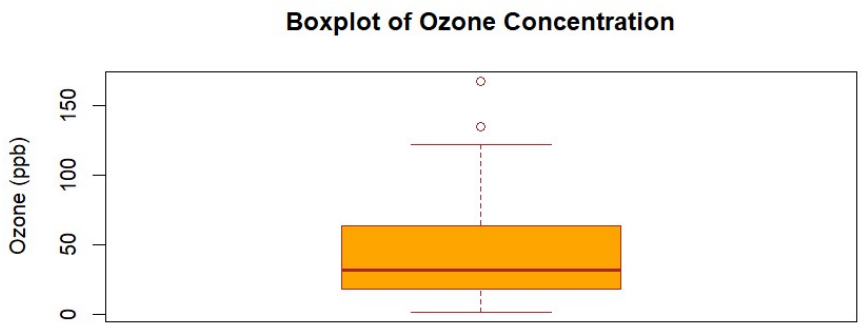
```
# Step 1: Load the dataset
data("airquality")

# Step 2: View the first few rows
head(airquality)

# Step 3: Summary of the dataset
summary(airquality)

# Step 4: Basic histogram to visualize Ozone concentration
hist(airquality$Ozone,
     main = "Ozone Concentration in Air",
     xlab = "Ozone (ppb)",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")

# Step 5: Boxplot to check ozone concentration spread and outliers
boxplot(airquality$Ozone,
        main = "Boxplot of Ozone Concentration",
        ylab = "Ozone (ppb)",
        col = "orange",
        border = "brown")
```

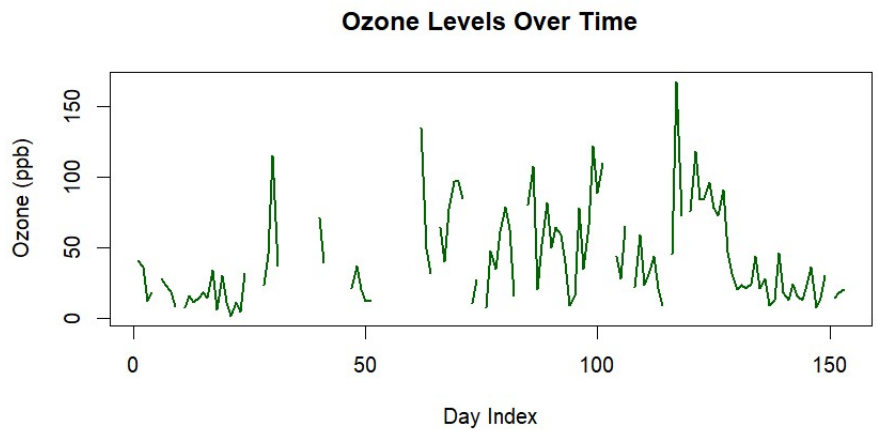
```
# Step 6: Line plot for Ozone levels over time (days)
plot(airquality$Ozone,
     type = "l",
     main = "Ozone Levels Over Time",
     xlab = "Day Index",
     ylab = "Ozone (ppb)",
     col = "darkgreen",
     lwd = 2)

# Add points to highlight data
points(airquality$Ozone, col = "blue", pch = 16)
```

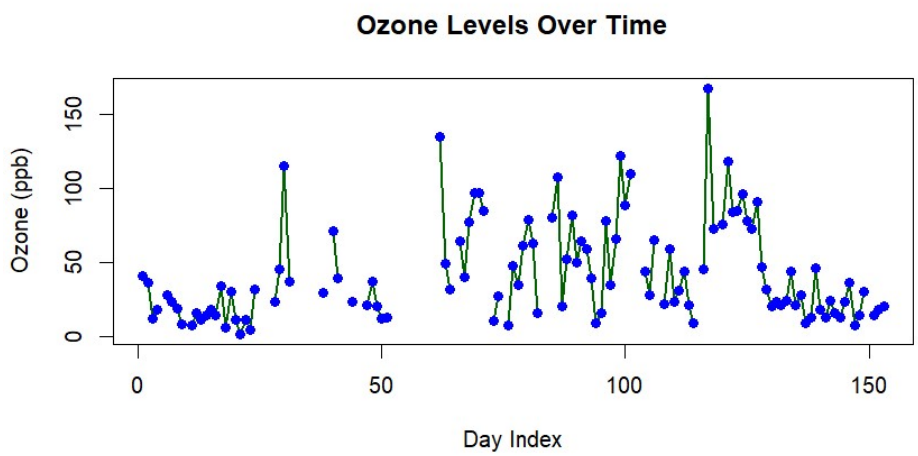
OUTPUT:

```
Ozone Solar.R Wind Temp Month Day
1    41     190  7.4   67     5   1
2    36     118  8.0   72     5   2
3    12     149 12.6   74     5   3
4    18     313 11.5   62     5   4
5    NA      NA 14.3   56     5   5
6    28      NA 14.9   66     5   6

 Ozone            Solar.R          Wind            Temp          Month
Min.   :  1.00   Min.   :  7.0   Min.   : 1.700   Min.   :56.00   Min.   :5.000
1st Qu.: 18.00   1st Qu.:115.8   1st Qu.: 7.400   1st Qu.:72.00   1st Qu.:6.000
Median : 31.50   Median :205.0   Median : 9.700   Median :79.00   Median :7.000
Mean   : 42.13   Mean   :185.9   Mean   : 9.958   Mean   :77.88   Mean   :6.993
3rd Qu.: 63.25   3rd Qu.:258.8   3rd Qu.:11.500   3rd Qu.:85.00   3rd Qu.:8.000
Max.   :168.00   Max.   :334.0   Max.   :20.700   Max.   :97.00   Max.   :9.000
NA's   :37       NA's   :7
    Day
Min.   : 1.0
1st Qu.: 8.0
Median :16.0
Mean   :15.8
```




Boxplot of Ozone Concentration
Ozone (ppb)
0
50
100
150


Ozone Levels Over Time
Ozone (ppb)
0
50
100
150
0
50
100
150
Day Index


Ozone Levels Over Time
Ozone (ppb)
0
50
100
150
0
50
100
150
Day Index

WEEK – 4

4. a) Load iris dataset in R, visualize the data using different Box plots including group by option and also use color palette to represent species.

AIM:

To visualize the iris data using different Box plots including group and also use color palette to represent species.

DESCRIPTION:

The iris dataset is loaded into the R environment. A summary of the dataset is displayed, showing statistical information such as minimum, maximum, median, and quartiles for each numeric variable. A boxplot of **Sepal Length** grouped by species is created. Different colors (lightblue, salmon, lightgreen) are used to distinguish the three species, and a dark blue border is applied to the boxes. A second boxplot is generated for **Petal Width** grouped by species, again using a different set of custom colors (skyblue, orange, lightpink) to highlight the differences among species. A color palette is generated using the rainbow() function to create three distinct colors. A boxplot of **Sepal Width** grouped by species is drawn using the rainbow color palette. This helps visualize how Sepal Width varies among the three iris species.

Overall, the program uses multiple boxplots and different color schemes to compare the flower measurements across species.

PROGRAM:

```
# Step 1: Load iris dataset
data("iris")

# Step 2: View dataset summary
summary(iris)

# Step 3: Boxplot of Sepal Length grouped by Species with color
boxplot(Sepal.Length ~ Species,
        data = iris,
        main = "Sepal Length by Species",
        xlab = "Species",
        ylab = "Sepal Length (cm)",
        col = c("lightblue", "salmon", "lightgreen"),
        border = "darkblue")

# Step 4: Boxplot of Petal Width grouped by Species with different colors
boxplot(Petal.Width ~ Species,
```

```
      data = iris,
      main = "Petal Width by Species",
      xlab = "Species",
      ylab = "Petal Width (cm)",
      col = c("skyblue", "orange", "lightpink"),
      border = "black")
```

 # Step 5: Use RColorBrewer for color palette

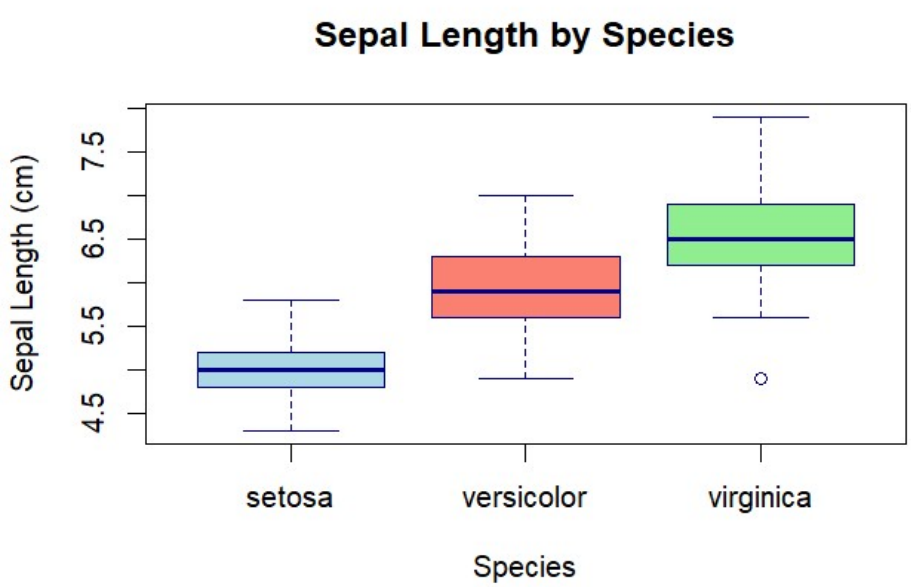
```
palette <- rainbow(3)
```

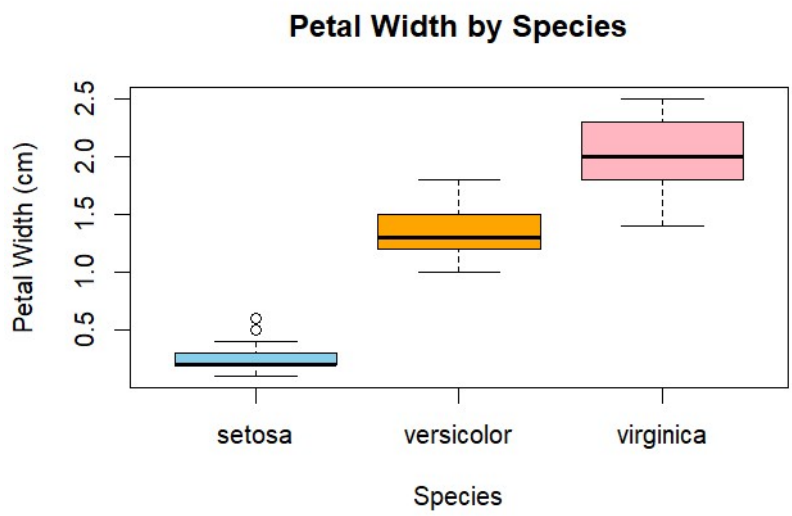
 # Step 6: Boxplot for Sepal Width grouped by Species using palette

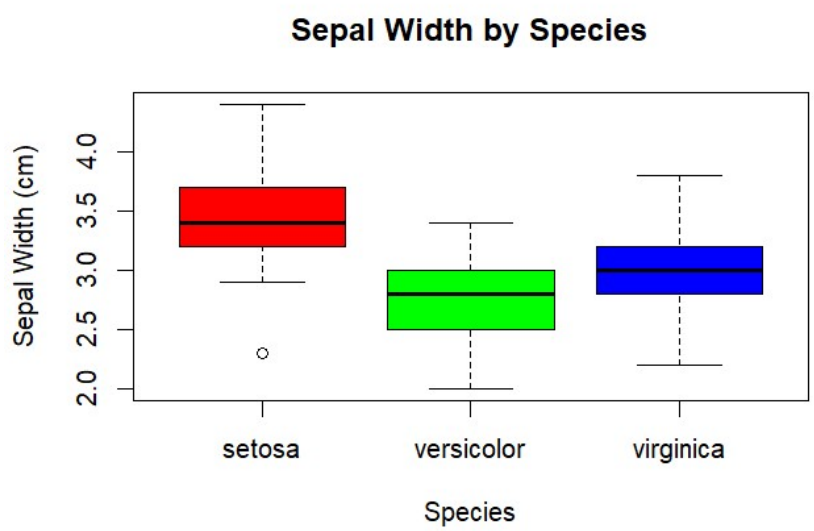
```
boxplot(Sepal.Width ~ Species,
      data = iris,
      main = "Sepal Width by Species",
      col = palette,
      xlab = "Species",
      ylab = "Sepal Width (cm)")
```

OUTPUT:

```
 Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
Median :5.800   Median :3.000   Median :4.350   Median :1.300
Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
       Species
setosa    :50
versicolor:50
virginica :50
```




Petal Width by Species
Petal Width (cm)
0.5
1.0
1.5
2.0
2.5
setosa
versicolor
virginica
Species


Sepal Width by Species
Sepal Width (cm)
2.0
2.5
3.0
3.5
4.0
setosa
versicolor
virginica
Species

b) Load air quality dataset in R and visualize air quality parameters using box plots.

AIM:

To visualize air quality parameters on air quality dataset using box plots.

DESCRIPTION:

The program analyzes and visualizes the airquality dataset using boxplots in R. The variables include Ozone concentration, Solar Radiation, Wind speed, and Temperature. The built-in airquality dataset is loaded into R. A summary of the dataset is displayed, showing statistical details such as minimum, median, mean, and maximum values for each variable. This provides an initial understanding of the data distribution.

Four separate boxplots are created for Ozone, Solar Radiation, Wind speed, and Temperature. **par(mfrow = c(2,2))** is used to divide the plotting area into a 2 × 2 grid, allowing four boxplots to be displayed together in one window for easy comparison. Each variable is plotted with a different color to visually distinguish them. Ozone → light blue, Solar Radiation → light green, Wind → light yellow, Temperature → light pink. The boxplots show the spread of data, median values, and presence of any outliers for each parameter.

Overall, the program visually summarizes the variability in air quality parameters, helps identify outliers, and allows comparison of the distribution of Ozone, Solar Radiation, Wind, and Temperature across the dataset.

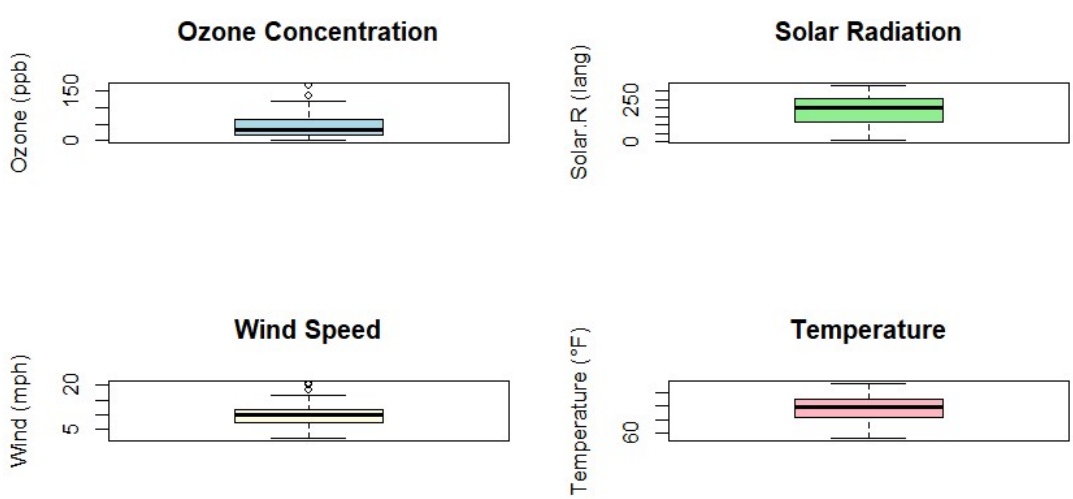
PROGRAM:

```
# Step 1: Load airquality dataset
data("airquality")

# Step 2: View dataset summary
summary(airquality)

# Step 3: Boxplots for Ozone, Solar.R, Wind, and Temp
par(mfrow = c(2, 2))  # 2x2 layout

boxplot(airquality$Ozone,
        main = "Ozone Concentration",
        ylab = "Ozone (ppb)",
        col = "lightblue")

boxplot(airquality$Solar.R,
        main = "Solar Radiation",
        ylab = "Solar.R (lang)",
        col = "lightgreen")
```

```
boxplot(airquality$Wind,
        main = "Wind Speed",
        ylab = "Wind (mph)",
        col = "lightyellow")

boxplot(airquality$Temp,
        main = "Temperature",
        ylab = "Temperature (°F)",
        col = "lightpink")

par(mfrow = c(1, 1))  # Reset layout
```

OUTPUT:

```
     Ozone           Solar.R           Wind             Temp
 Min.   :  1.00   Min.   :  7.0   Min.   : 1.700   Min.   :56.00
 1st Qu.: 18.00   1st Qu.:115.8   1st Qu.: 7.400   1st Qu.:72.00
 Median : 31.50   Median :205.0   Median : 9.700   Median :79.00
 Mean   : 42.13   Mean   :185.9   Mean   : 9.958   Mean   :77.88
 3rd Qu.: 63.25   3rd Qu.:258.8   3rd Qu.:11.500   3rd Qu.:85.00
 Max.   :168.00   Max.   :334.0   Max.   :20.700   Max.   :97.00
 NA's   :37       NA's   :7
     Month            Day
 Min.   :5.000   Min.   : 1.0
 1st Qu.:6.000   1st Qu.: 8.0
 Median :7.000   Median :16.0
 Mean   :6.993   Mean   :15.8
 3rd Qu.:8.000   3rd Qu.:23.0
 Max.   :9.000   Max.   :31.0
```



WEEK – 5

5. Visualize iris dataset using simple scatter, multivariate scatter plot and also visualize scatter plot matrix to visualize multiple variables across each other.

AIM:

To visualize iris dataset using simple scatter, multivariate scatter plot and visualize multiple variables across each other.

DESCRIPTION:

The program loads the built-in iris dataset and visualizes it using different scatter plot techniques. Initially, the dataset is displayed to observe its structure and values. A simple scatter plot is created between Sepal Length and Sepal Width, where each data point represents a flower and colors are used to distinguish different species. A legend is added to improve clarity and interpretation of the species-wise distribution.

Next, a multivariate scatter plot is drawn between Sepal Length and Petal Length, again using color to represent species, which helps in comparing relationships across multiple variables simultaneously. Finally, a scatter plot matrix is generated for all four numerical attributes of the iris dataset, allowing observation of pairwise relationships and patterns among variables. Overall, the program helps in identifying trends, correlations, and species separation through effective graphical visualization.

PROGRAM:

```
# Load iris dataset
data("iris")
print(iris)

# Simple scatter plot: Sepal Length vs Sepal Width
plot(iris$Sepal.Length, iris$Sepal.Width,
    main = "Scatter Plot: Sepal Length vs Sepal Width",
    xlab = "Sepal Length (cm)",
    ylab = "Sepal Width (cm)",
    col = iris$Species,
    pch = 19)

# Add legend
legend("topright", legend = levels(iris$Species),
       col = 1:3, pch = 19)

# Multivariate scatter plot: Sepal Length vs Petal Length, colored by Species
plot(iris$Sepal.Length, iris$Petal.Length,
```

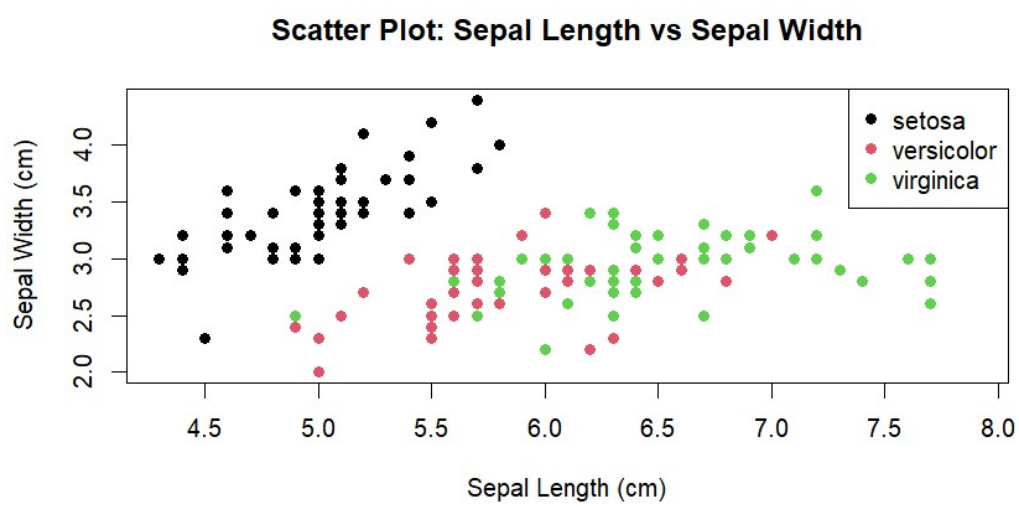
```
    main = "Multivariate Scatter: Sepal vs Petal Length",
    xlab = "Sepal Length (cm)",
    ylab = "Petal Length (cm)",
    col = iris$Species,
    pch = 19)
```

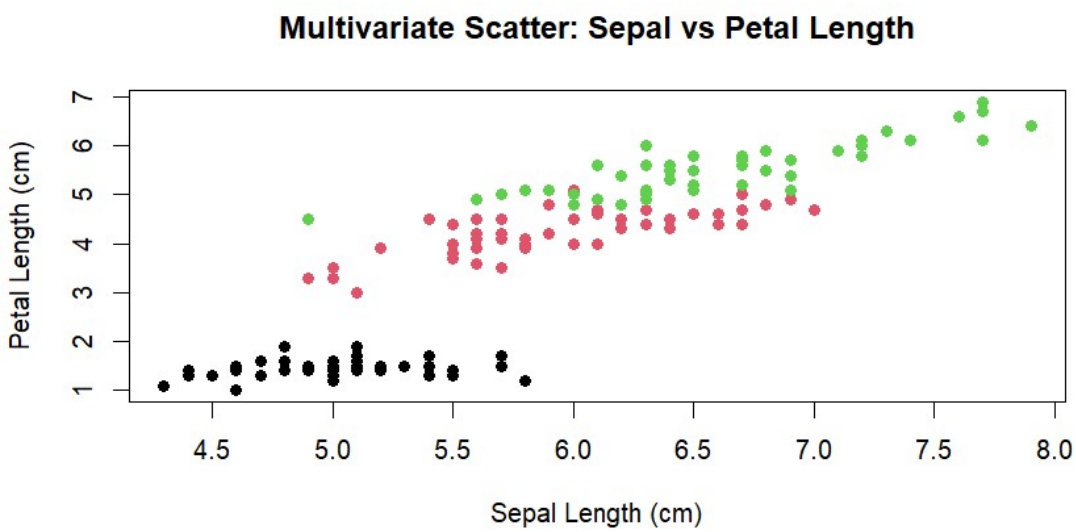
 # Scatter plot matrix

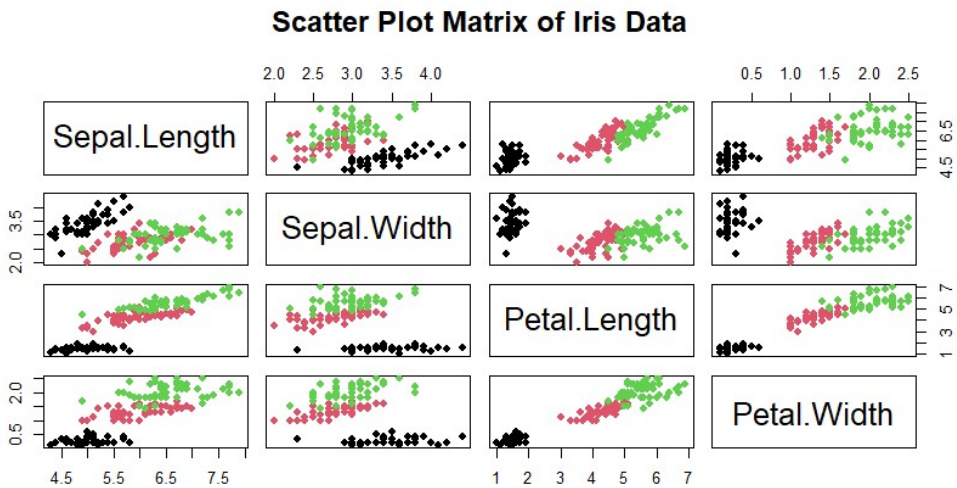
```
pairs(iris[1:4],
     main = "Scatter Plot Matrix of Iris Data",
     col = iris$Species,
     pch = 19)
```

OUTPUT:

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```




Multivariate Scatter: Sepal vs Petal Length
Petal Length (cm)
Sepal Length (cm)
1
2
3
4
5
6
7
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0


Scatter Plot Matrix of Iris Data
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width
2.0 2.5 3.0 3.5 4.0
0.5 1.0 1.5 2.0 2.5
4.5 6.5
2.0 3.5
1 3 5 7
0.5 2.0
4.5 5.5 6.5 7.5
1 2 3 4 5 6 7

WEEK – 6

6. Load diamonds dataset in R and visualize the structure in datasets with large data points using hexagon binning and also add color palette.

AIM:

To visualize the structure of diamonds dataset with large data points using hexagon binning.

DESCRIPTION:

The program demonstrates the use of hexbin plots and mosaic plots in R for visualizing large datasets and categorical data. Required libraries ggplot2, hexbin, and vcd are loaded to support these visualizations. Using the diamonds dataset, a hexbin plot is created to analyze the relationship between carat and price, where hexagonal bins represent the density of data points and color intensity indicates concentration. This helps in identifying trends and reducing overplotting in dense data.

Next, the built-in HairEyeColor dataset is converted into a data frame and a mosaic plot is generated to visualize the association between hair color and eye color. The area of each tile represents the frequency of each hair–eye color combination. Overall, the program effectively illustrates how different visualization techniques can be used to explore numerical and categorical data patterns.

PROGRAM:

```
# Install and Load required packages
install.packages("hexbin")
install.packages("vcd")
library(ggplot2)
library(hexbin)
library(vcd)  # For mosaic plot

# Load diamonds dataset from ggplot2
data("diamonds")

# Hexbin plot: carat vs price
ggplot(diamonds, aes(x = carat, y = price)) +
  stat_binhex(bins = 50) +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(title = "Hexbin Plot: Carat vs Price",
       x = "Carat",
       y = "Price")

# Load HairEyeColor dataset (built-in)
data("HairEyeColor")
```

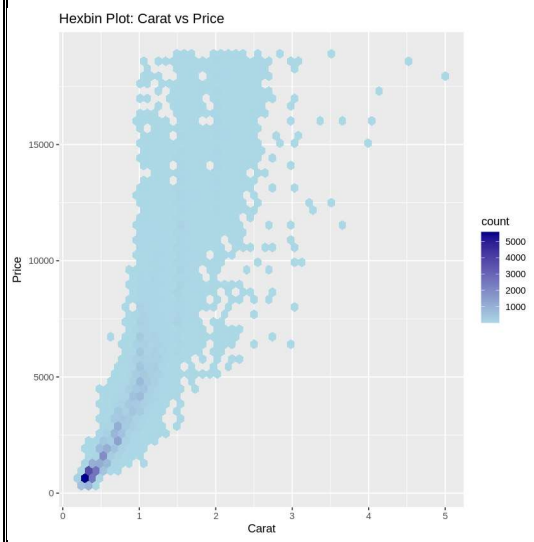
 # Convert to data frame

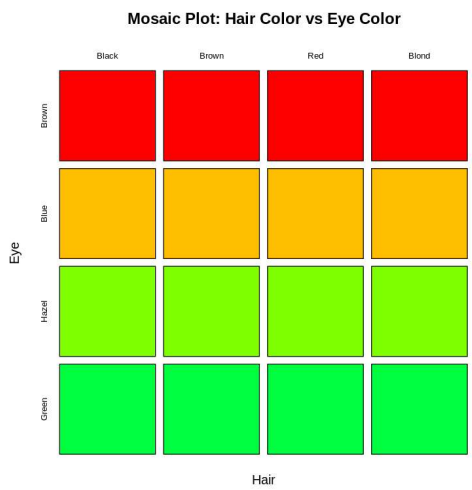
```
hair_eye_df <- as.data.frame(HairEyeColor)
```

 # Mosaic plot for categorical data

```
mosaicplot(~ Hair + Eye,
        data = hair_eye_df,
        main = "Mosaic Plot: Hair Color vs Eye Color",
        col = rainbow(8),
        border = "black")
```

OUTPUT:





WEEK – 7

7. Load Hair Eye Color dataset in R and plot categorical data using mosaic plot.

AIM:

To Load Hair Eye Color dataset in R and plot categorical data using mosaic plot.

DESCRIPTION:

The program uses the built-in HairEyeColor dataset in R, which represents the frequency distribution of hair color, eye color, and gender. The dataset is first viewed in its original table format. It is then optionally converted into a data frame (hair_eye_df) for easier manipulation or analysis.

A **mosaic plot** is created to visualize the relationship between **hair color** and **eye color**. The mosaic plot displays the proportion of individuals for each combination of hair and eye color, where the size of each rectangle is proportional to the frequency of that combination. Colors are used to enhance the visualization, making patterns and associations between hair and eye colors easier to interpret.

PROGRAM:

```
# Load the built-in HairEyeColor dataset
data("HairEyeColor")

# View the dataset
HairEyeColor

# Convert the table to a data frame (optional)
hair_eye_df <- as.data.frame(HairEyeColor)

# Create a mosaic plot
mosaicplot(HairEyeColor,
        main = "Mosaic Plot of Hair Color vs Eye Color",
        xlab = "Hair Color",
        ylab = "Eye Color",
        color = TRUE)
```

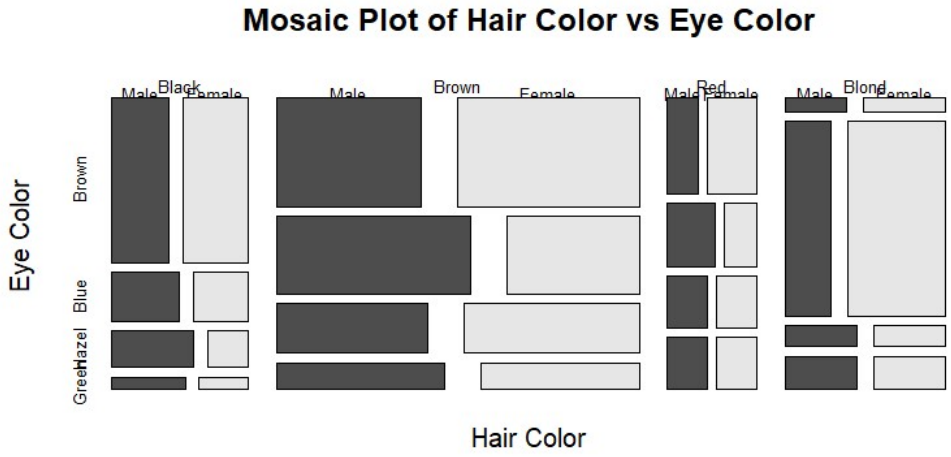
OUTPUT:

Sex = Male

Hair	Eye Brown	Blue	Hazel	Green
Black	32	11	10	3
Brown	53	50	25	15
Red	10	10	7	7
Blond	3	30	5	8

, , Sex = Female

Hair	Eye Brown	Blue	Hazel	Green
Black	36	9	5	2
Brown	66	34	29	14
Red	16	7	7	7
Blond	4	64	5	8



WEEK – 8

8. Load mtcars dataset in R and visualize data using heat map.

AIM:

To Load mtcars dataset in R and visualize data using heat map.

DESCRIPTION:

The program loads the built-in **mtcars** dataset and displays its summary to understand the basic statistics of each variable. The data is then **normalized using the scale() function** so that all variables are brought to a common scale, which improves heatmap interpretation. A **heatmap** is generated using base R to visualize patterns, similarities, and variations among cars and variables. The color intensity represents the magnitude of values, making it easier to identify clusters and relationships within the dataset.

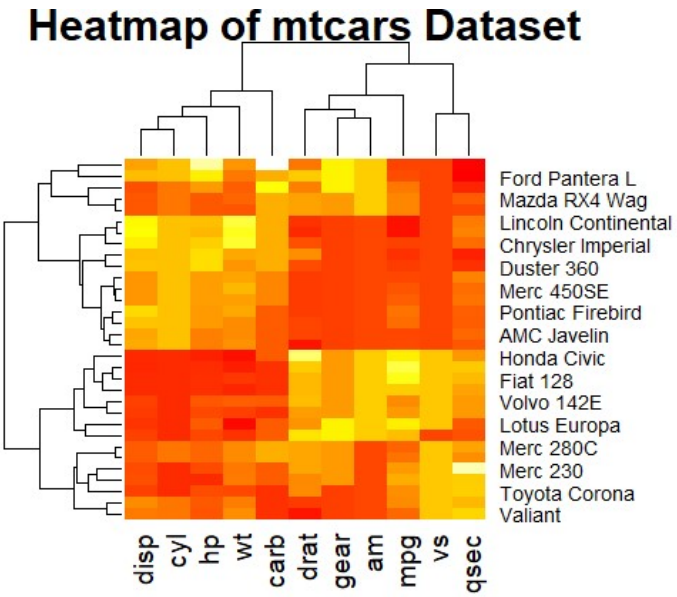
PROGRAM:

```
# Load the dataset
data("mtcars")

#Summary of dataset
summary(mtcars)

# Normalize the data (optional for better heatmap visualization)
mtcars_scaled <- scale(mtcars)

# Heatmap using base R
heatmap(mtcars_scaled,
        main = "Heatmap of mtcars Dataset",
        col = heat.colors(256),
        scale = "none")
```

OUTPUT:

```
     mpg             cyl             disp           hp              drat
 Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0   Min.   :2.760
 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5   1st Qu.:3.080
 Median :19.20   Median :6.000   Median :196.3   Median :123.0   Median :3.695
 Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7   Mean   :3.597
 3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0   3rd Qu.:3.920
 Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0   Max.   :4.930
```

```
      wt              qsec            vs                 am                gear
 Min.   :1.513   Min.   :14.50   Min.   :0.0000   Min.   :0.0000   Min.   :3.000
 1st Qu.:2.581   1st Qu.:16.89   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:3.000
 Median :3.325   Median :17.71   Median :0.0000   Median :0.0000   Median :4.000
 Mean   :3.217   Mean   :17.85   Mean   :0.4375   Mean   :0.4062   Mean   :3.688
 3rd Qu.:3.610   3rd Qu.:18.90   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:4.000
 Max.   :5.424   Max.   :22.90   Max.   :1.0000   Max.   :1.0000   Max.   :5.000
     carb
 Min.   :1.000
 1st Qu.:2.000
 Median :2.000
 Mean   :2.812
 3rd Qu.:4.000
 Max.   :8.000
```



WEEK – 9

9. Install leaflet library in R and perform different map visualizations.

AIM:

To install leaflet library and perform different map visualizations.

DESCRIPTION:

This R program demonstrates the use of the leaflet package to create interactive geographic visualizations. Leaflet enables the visualization of spatial data on web-based maps with zooming, panning, and popup features.

First, the leaflet library is loaded after installing it (installation required only once). A basic interactive map is created using leaflet() and addTiles(), which adds the default OpenStreetMap tiles. A marker is placed at the geographic coordinates of Bangalore, and a popup is attached to display the city name when the marker is clicked.

Next, the program illustrates how to add circle markers to represent locations with an area of influence. A circle is drawn at the coordinates of Delhi with a specified radius and color, along with a popup label. This shows how geographic points can be visually emphasized using circles instead of simple markers.

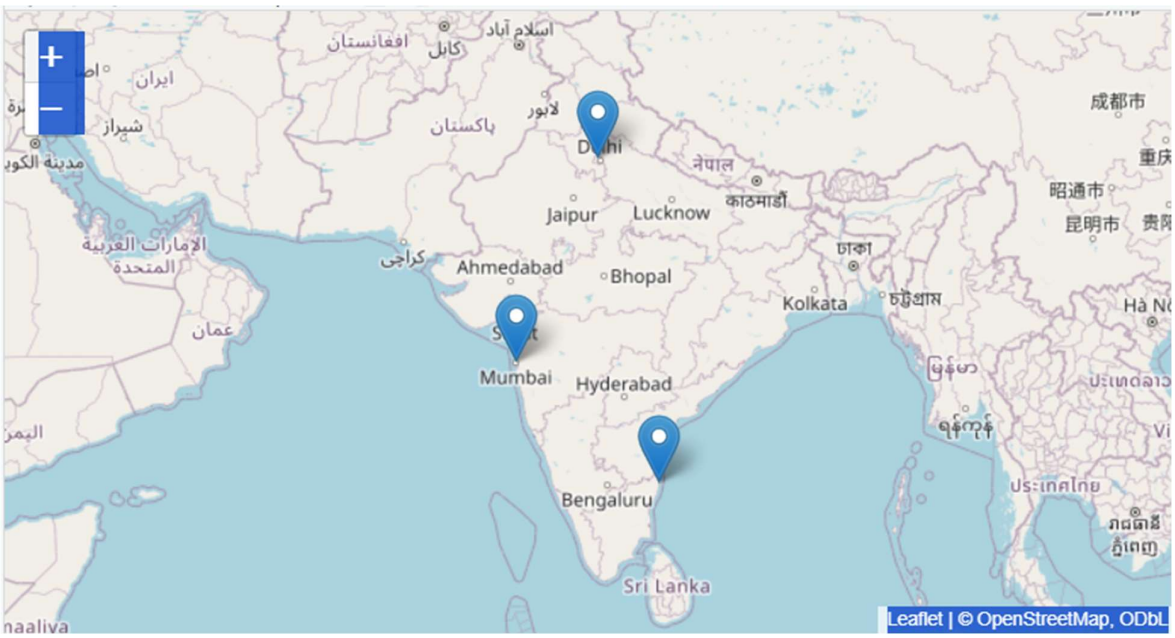
Finally, the program demonstrates plotting multiple locations on a single map. A data frame containing the names and coordinates of Delhi, Mumbai, and Chennai is created. Using this data frame, multiple markers are added simultaneously to the map, with popups displaying the respective city names.

PROGRAM:

```
# Install leaflet (only once)
# install.packages("leaflet")

# Load leaflet library
library(leaflet)

# Basic Map with default view
leaflet() %>%
  addTiles() %>%
  addMarkers(lng = 77.5946, lat = 12.9716, popup = "Bangalore")
# Map with circles
leaflet() %>%
  addTiles() %>%
  addCircles(lng = 77.1025, lat = 28.7041, radius = 50000, color = "blue", popup =
             "Delhi")
```

```
# Multiple points
cities <- data.frame(
  name = c("Delhi", "Mumbai", "Chennai"),
  lat = c(28.7041, 19.0760, 13.0827),
  lng = c(77.1025, 72.8777, 80.2707)
)

leaflet(cities) %>%
  addTiles() %>%
  addMarkers(~lng, ~lat, popup = ~name)
```

OUTPUT:



WEEK – 10

10. Visualize iris dataset using 3d graphs such as scatter3d, cloud, xyplot.

AIM:

To Visualize iris dataset using 3d graphs such as scatter3d, cloud, xyplot.

DESCRIPTION:

This R program demonstrates different techniques for 3D data visualization using the Iris dataset. It employs multiple R libraries to create three-dimensional scatter plots, helping to understand relationships among variables more effectively.

First, the required libraries scatterplot3d, lattice, and car are loaded. Each library provides a different approach to visualizing multivariate data in three dimensions.In the first part, a 3D scatter plot is created using the scatterplot3d package. The variables Sepal Length, Petal Length, and Sepal Width are plotted on the X, Y, and Z axes respectively. Different colors are used to distinguish iris species, allowing visual comparison of class separation in three-dimensional space.

Next, a cloud plot is generated using the lattice package. This plot visualizes the same three variables in a 3D layout, grouped by species. The auto.key = TRUE option provides a legend, making it easier to identify different species clusters.

Finally, the program uses the scatter3d function from the car package to create an interactive 3D scatter plot. This plot allows rotation and exploration of data points and displays separate panels for each iris species, improving clarity in comparative analysis.

PROGRAM:

```
# Install required libraries
# install.packages("scatterplot3d")
#install.packages("lattice")
#install.packages("rgl")
#install.packages("car")

library(scatterplot3d)
library(lattice)
library(car)

# 1. 3D Scatter Plot using scatterplot3d
scatterplot3d(iris$Sepal.Length, iris$Petal.Length, iris$Sepal.Width,
        color = as.numeric(iris$Species),
        pch = 19,
        main = "3D Scatter Plot of Iris",
        xlab = "Sepal Length",
```

```
            ylab = "Petal Length",
            zlab = "Sepal Width")
```

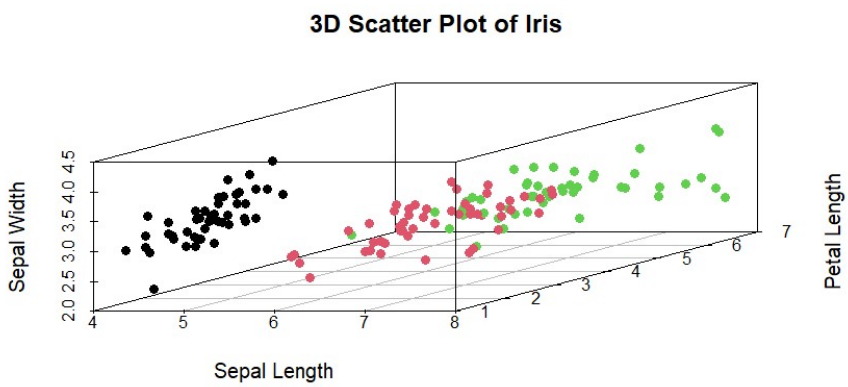
 # 2. Cloud plot using lattice

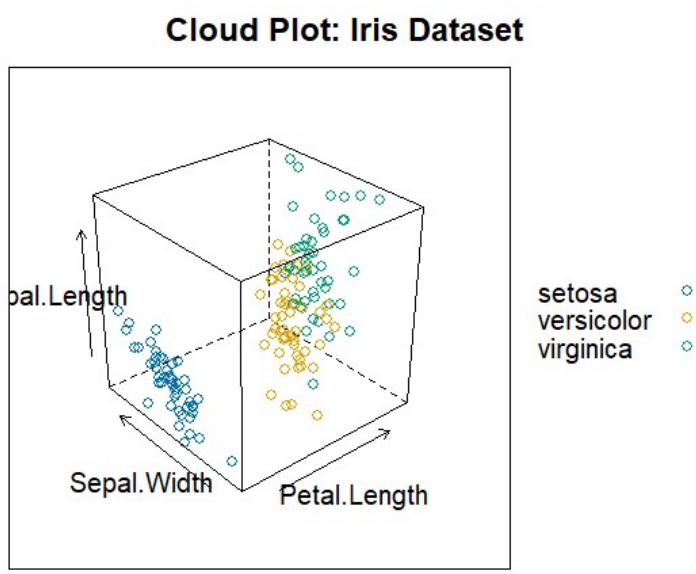
```
cloud(Sepal.Length ~ Petal.Length * Sepal.Width,
      data = iris,
      groups = iris$Species,
      auto.key = TRUE,   main = "Cloud Plot: Iris Dataset")
```

 # 3. scatter3d from car package (interactive 3D plot)

```
scatter3d(Sepal.Length ~ Petal.Length + Sepal.Width | Species,
          data = iris,
          surface = FALSE,
          grid = FALSE)
```

OUTPUT:





WEEK – 11

11. Make use of correlogram to visualize data in correlation matrices for iris dataset.

AIM:

To visualize data in correlation matrices for iris dataset.

DESCRIPTION:

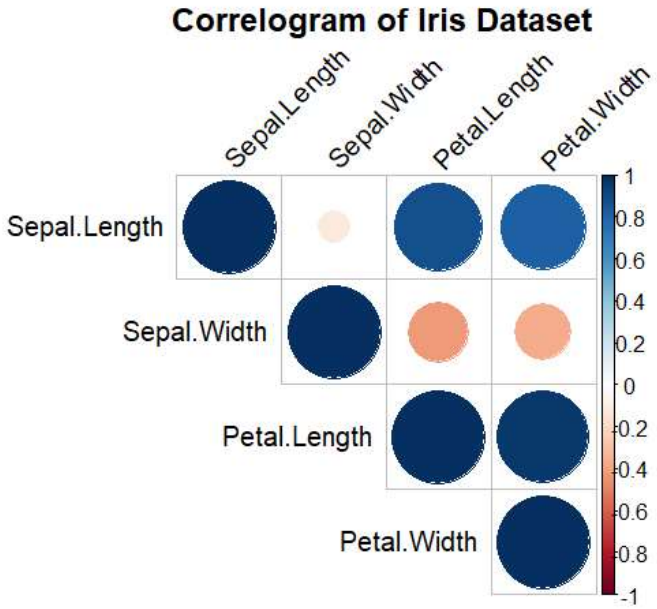
The program uses the corrplot package in R to visualize the correlation matrix of the Iris dataset. First, only the numerical attributes (Sepal Length, Sepal Width, Petal Length, and Petal Width) are selected. The correlation matrix is then computed using the cor() function.

The correlogram visually represents the strength and direction of correlations using circular markers. Larger and darker circles indicate stronger correlations, while smaller circles represent weaker relationships. This visualization helps in quickly identifying strong positive or negative correlations among variables, making it useful for exploratory data analysis and feature selection.

PROGRAM:

```
# Install required package (run only once)
# install.packages("corrplot")

# Load the corrplot library
library(corrplot)

# Load iris dataset
data(iris)

# Select only numerical columns
iris_numeric <- iris[, 1:4]

# Compute correlation matrix
cor_matrix <- cor(iris_numeric)

# Display correlogram
corrplot(cor_matrix,
        method = "circle",      # Visualization method
        type = "upper",         # Show upper triangle
        tl.col = "black",       # Text label color
        tl.srt = 45,            # Text label rotation
        title = "Correlogram of Iris Dataset",
        mar = c(0,0,2,0))
```

OUTPUT:


Correlogram of Iris Dataset
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

WEEK – 12

12. Install maps library in R and draw different map visualizations.

AIM:

To install maps library and draw different map visualizations.

DESCRIPTION:

The program demonstrates the use of the **maps** and **mapdata** packages in R to visualize geographic data. A **world map** is first displayed using filled polygons with customized background and colors. The program then highlights **India** on the world map by specifying it as a region.

A **state-level map of the USA** is generated using the built-in state map database. Different colors are applied to US states to improve visual distinction. Selected **Asian countries** are combined to represent a map of Asia. This approach overcomes the limitation of continent-level mapping in the maps package.

The program next redraws the **map of India** with a neutral color for clarity. Major Indian cities are marked using **latitude and longitude points**. City names are added as text labels, demonstrating annotation on maps.

PROGRAM:

```
# Step 1: Install and load required packages
#install.packages("maps")
#install.packages("mapdata")

library(maps)
library(mapdata)

# Step 2: World map
map("world",
   fill = TRUE,
   col = "lightblue",
   bg = "lightyellow",
   main = "World Map")

# Step 3: Map of India
map("world", regions = "India",
   fill = TRUE,
   col = "orange",
   main = "Map of India")

# Step 4: Map of USA (states)
map("state",
   fill = TRUE,
```

```
    col = rainbow(50),
    main = "Map of USA States")
```

 # Step 5: Map of Asia

```
asia_countries <- c(
  "India", "China", "Japan", "Pakistan", "Bangladesh", "Sri Lanka",
  "Nepal", "Bhutan", "Myanmar", "Thailand", "Malaysia", "Indonesia",
  "Philippines", "Vietnam", "South Korea", "North Korea"
)

map("world",
    regions = asia_countries,
    fill = TRUE,
    col = "lightgreen",
    main = "Map of Asia (Selected Countries)")
```

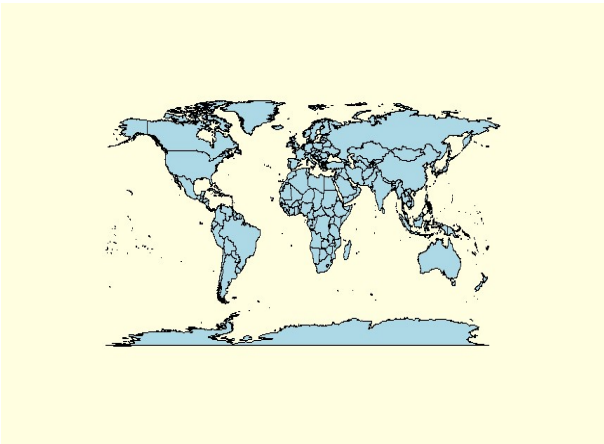
 # Step 6: Add points to a map (e.g., major cities in India)

```
map("world", regions = "India",
    fill = TRUE,
    col = "lightgray",
    main = "India Map with Cities")
```

 # Add city points (lat, long)

```
points(x = c(77.5946, 72.8777, 80.2707),  # Longitudes
       y = c(12.9716, 19.0760, 13.0827),  # Latitudes
       col = "red", pch = 19)

text(x = c(77.5946, 72.8777, 80.2707),
     y = c(12.9716, 19.0760, 13.0827),
     labels = c("Bangalore", "Mumbai", "Chennai"),
     pos = 4, cex = 0.8)
```

OUTPUT:



Mumbai